

[6353]-162

T.E. (Mechanical)/(Mechanical Sandwich)
DESIGN OF MACHINE ELEMENTS
(2019 Pattern) (Semester - I) (302043)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

Q1) a) Explain self-locking and overhauling conditions of power screw. **[6]**

- b) A machine vice, as shown in Fig. 1 has single-start, square threads with 22 mm nominal diameter and 5 mm pitch. The outer and inner diameters of the friction collar are 55 and 45 mm respectively. The coefficients of friction for thread and collar are 0.15 and 0.17 respectively. The machinist can comfortably exert a force of 125 N on the handle at a mean radius of 150 mm. Assuming uniform wear for the collar, calculate; **[12]**

- i) the clamping force developed between the jaws; and
- ii) the overall efficiency of the clamp.

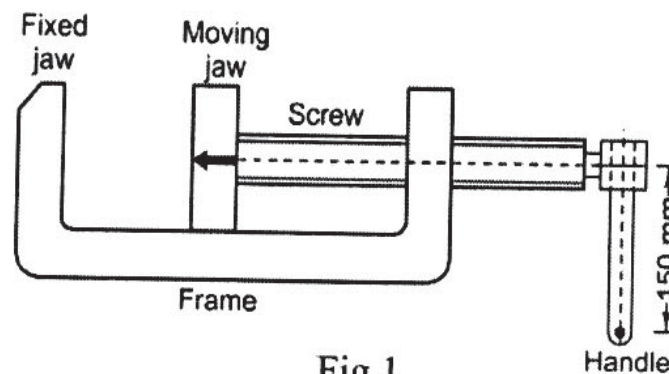


Fig.1

OR

P.T.O.

Q2) a) Derive an expression for efficiency of self-locking screw for square thread. [6]

b) The nominal diameter of a triple-threaded square screw is 50 mm, while the pitch is 8 mm. It is used with a collar having an outer diameter of 100 mm and inner diameter as 65 mm. The coefficient of friction at the thread surface as well as at the collar surface can be taken as 0.15. The screw is used to raise a load of 15 kN. Using the uniform wear theory for collar friction, Calculate: [12]

- torque required to raise the load;
- torque required to lower the load; and
- the force required to raise the load, if applied at a radius of 500 mm.

Q3) a) Explain with neat sketch Modified Goodman diagram. [5]

b) A forged steel bar, 50 mm in diameter, is subjected to a reversed bending stress of 250 N/mm^2 . The bar is made of steel 40C8 ($S_{ut} = 600 \text{ N/mm}^2$). Calculate the life of the bar for a reliability of 90%. Assume Surface finish factor $K_a = 0.44$, For 50 mm diameter, $K_b = 0.85$ For 90% reliability, $K_c = 0.897$. [12]

OR

Q4) a) Draw time- stress curves for fluctuating stress, completely reversed stress. [5]

b) A cantilever beam made of cold drawn steel 20C8 ($S_{ut} = 540 \text{ N/mm}^2$) is subjected to a completely reversed load of 1000 N as shown in Fig 2. The notch sensitivity factor 'q' at the fillet can be taken as 0.85 and the expected reliability is 90%. Determine the diameter 'd' of the beam for a life of 10000 cycles. Assume surface finish, size and reliability factors are 0.78, 0.85 and 0.897 respectively. Also assume $K_t = 1.35$. [12]

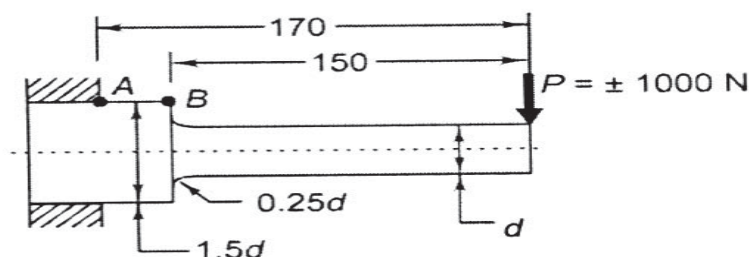


Fig 2.

- Q5) a)** Write a note on bolt of uniform strength and explain the methods for achieving bolts of uniform strength. [5]
- b)** A steel plate subjected to a force of 5 kN and fixed to a channel by means of three identical bolts is shown in Fig.3. The bolts are made from plain carbon steel 45C8 ($S_{yt} = 380 \text{ N/mm}^2$) and the factor of safety is 3. Specify the size of bolts. [12]

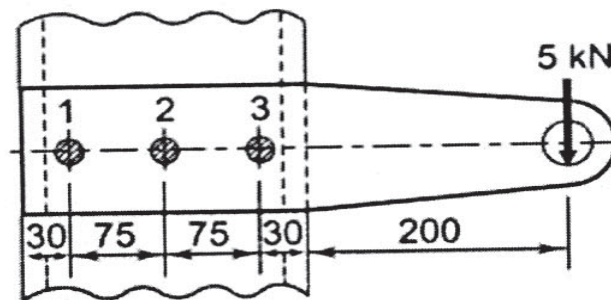


Fig.3

OR

- Q6) a)** Explain with suitable sketch the type of fillet joints. [5]
- b)** A steel plate, 100mm wide and 10 mm thick, is joined with another steel plate by means of single transverse and double parallel fillet welds, as shown in Fig.4. The strength of the welded joint should be equal to the strength of the plates to be joined. The permissible tensile and shear stresses for the weld material and the plates are 70 and 50 N/mm^2 respectively. Find the length of each parallel fillet weld. Assume the tensile force acting on the plates as static. [12]

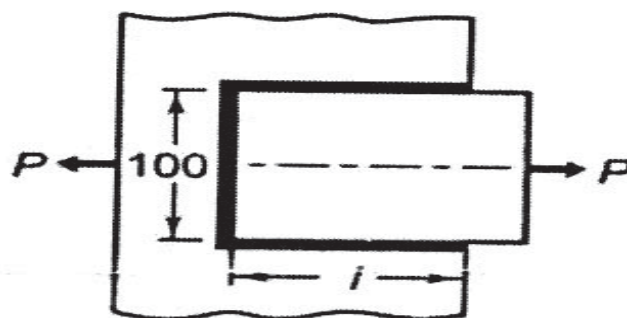


Fig.4

- Q7) a)** Explain the following terms related to the helical compression spring along with the equations required to calculate it. **[6]**
- i) Solid Length
 - ii) Compressed length
 - iii) Free Length
- b)** Derive load- stress and load- deflection equations for helical compression spring. **[12]**

OR

- Q8) a)** Explain resultant stiffness of springs for parallel and series connections. **[6]**
- b)** A helical compression spring, made of circular wire, is subjected to an axial force, which varies from 2.5 kN to 3.5 kN. Over this range of force, the deflection of the spring should be approximately 5 mm. The spring index can be taken as 5. The spring has square and ground ends. The spring is made of patented and cold-drawn steel wire with ultimate tensile strength of 1050 N/mm² and modulus of rigidity of 81370 N/mm². The permissible shear stress for the spring wire should be taken as 50% of the ultimate tensile strength **[12]**

Design the spring and calculate;

- i) wire diameter
- ii) mean coil diameter
- iii) number of active coils
- iv) total number of coils
- v) solid length of the spring
- vi) free length of the spring
- vii) required spring rate.

* * *

[6262]-164

T.E. (Mechanical Sandwich) (Mechanical)
DESIGN OF MACHINE ELEMENTS
(2019 Pattern) (Semester - I) (302043)

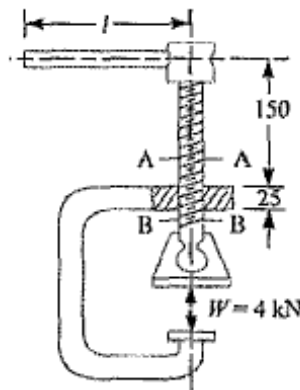
Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1) a)** A C-clamp, as shown in Figure, has trapezoidal threads of 12 mm outside diameter and 2mm pitch. The coefficient of friction for screw threads is 0.12 and for the collar is 0.25. The mean radius of the collar is 6 mm. If the force exerted by the operator at the end of the handle is 80N, find: 1) The length of handle; 2) The maximum shear stress in the body of the screw and where does this exist; and 3) The bearing pressure on the threads. **[8]**



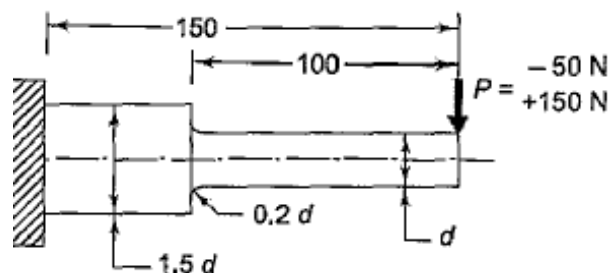
- b) Derive expression for torque required to raise the loads in case of square threads. **[5]**
- c) Prove that efficiency of self-locking square threads is less than 50%. **[4]**

OR

P.T.O.

- Q2) a)** A power screw having double start square threads of 25 mm nominal diameter and 5 mm pitch is acted upon by an axial load of 10 kN. The outer and inner diameters of screw collar are 50 mm and 20 mm respectively. The coefficient of thread friction and collar friction may be assumed as 0.2 and 0.15 respectively. The screw rotates at 12 r.p.m. Assuming uniform wear condition at the collar and allowable thread bearing pressure of 5.8 N/mm^2 , find: 1. the torque required to rotate the screw; 2. the stress in the screw; and 3. the number of threads of nut in engagement with screw. Also state the condition of screw. [8]
- b) Explain with neat sketch, differential screw and Compound screw. [4]
- c) Explain with neat sketch, re-circulating ball screw. State its application.[5]

- Q3) a)** A cantilever beam made of cold drawn steel 40C8 ($S_{ut} = 600 \text{ N/mm}^2$ and $S_{yt} = 380 \text{ N/mm}^2$) is shown in Figure. The force P acting at the free end varies from -50 N to $+150 \text{ N}$. The expected reliability is 90% and the factor of safety is 2. The notch sensitivity factor at the fillet is 0.9. Determine the diameter of the beam at the fillet cross-section. Take surface finish factor 0.77, Size factor 0.85, Reliability factor 0.897 and theoretical stress concentration factor 1.44. [8]



- b) What is stress concentration? What are the causes of stress concentration? [6]
- c) Define and Explain 1) Notch Sensitivity 2) Endurance limit. [4]

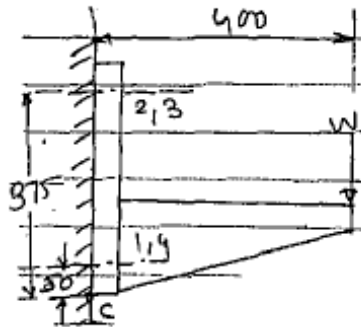
OR

- Q4) a)** A cantilever beam of circular cross section made of steel Fe 550 ($S_{ut} = 550 \text{ N/mm}^2$) is fixed at one end and subjected to a completely reversed load (P) of 15 kN is at the free end. The force P is perpendicular to the axis of beam. The distance between forces to fixed end is 200 mm. Take $K_t = 1.35$, $q = 0.85$, $K_a = 0.8$, $K_c = 0.897$ and the values of K_b is taken from table. The factor of safety is 2 Calculate the diameter of the beam for infinite life. [8]

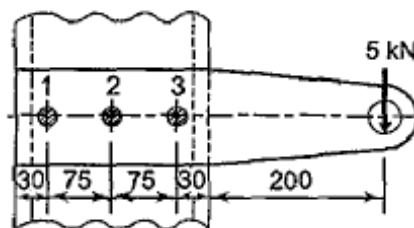
d (mm)	Kb
$d \leq 7.5$	1.0
$7.5 < d \leq 50$	0.85
$d > 50$	0.75

- b) Explain modified Goodman diagram. Draw neat labeled sketches of modified Goodman diagram for axial & Bending stresses. [5]
- c) Write a short note on Cumulative damage. [5]

- Q5) a)** A bracket shown in figure is fixed to steel column by means of four bolts of size M14. A load of W acts on the bracket. Take $\sigma_t = 84$ MPa. If ratio B/t for cross section of the arm bracket is 45. Determine 1) Maximum load on bracket, 2) Cross section of bolts. [8]



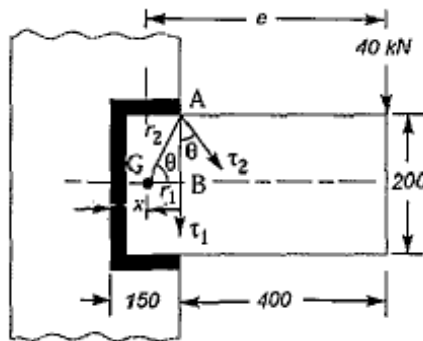
- b) A steel plate subjected to a force of 5 kN and fixed to a channel by means of three identical bolts is shown in Figure. The bolts are made from plain carbon steel 45C8 ($S_{yt} = 380$ N/mm²) and the factor of safety is 3. Specify the size of bolts. [5]



- c) Write a note on: Bolts of uniform strength. [5]

OR

- Q6) a)** A rectangular steel plate is welded as a cantilever to a vertical column and supports a single concentrated load 40 kN , as shown in Figure. Determine the Maximum shear stress if size of weld is 11.62 mm . **[8]**



- b) Discuss in brief strength of parallel fillet welds. **[5]**
- c) Prove that stress on the throat is equal to the ratio of force on weld to $0.707 \times s \times l$ **[5]**
- Q7) a)** A mechanism used in printing machinery consists of a tension spring assembled with a preload of 30 N . The wire diameter of spring is 2 mm with a spring index of 6 . The spring has 18 active coils. The spring wire is hard drawn and oil tempered having following material properties: Design shear stress = 680 MPa ; Modulus of rigidity = 80 kN/mm^2 Determine: 1) the initial torsional shear stress in the wire; 2) Spring rate; and 3) The force to cause the body of the spring to its yield strength. **[8]**
- b) Draw a neat sketch of a multi-leaf spring and show its essential parts. State functions of any two components. **[4]**
- c) What is mean by spring surge and what is its effect? **[5]**

OR

- Q8) a)** A safety valve of 60 mm diameter is to blow off at a pressure of 1.2 N/mm^2 . It is held on its seat by a close coiled helical spring. The maximum lift of the valve is 10 mm . Design a suitable compression spring of spring index 5 and providing an initial compression of 35 mm . The maximum shear stress in the material of the wire is limited to 500 MPa . The modulus of rigidity for the spring material is 80 kN/mm^2 . Calculate: 1) Diameter of the spring wire, 2) Mean coil diameter, 3) Number of active turns, and 4) Pitch of the coil. **[8]**
- b) Explain with the neat sketch, nipping of leaf spring. **[5]**
- c) Explain A.M. Wahl's factor and state its importance in the design of helical springs. **[4]**



Time : 3 Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answer Q.1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8, Q9 or Q10.*
- 2) *Neat diagram must be drawn wherever necessary.*
- 3) *Figures to the right indicate full marks.*
- 4) *Use of scientific calculators is allowed.*
- 5) *Assume suitable data if necessary.*

- Q1)** a) Explain the importance of preferred. Series in engineering and find numbers of R10 basic series from 1 to 10. **[4]**
- b) The frame of a hydraulic press consisting of two identical steel plates is shown in Fig.1. The maximum force P acting on the frame is 20 kN. The plates are made of steel 45C8 with tensile yield strength of 380 N/mm^2 . The factor of safety is 2.5. Determine plate thickness. **[6]**

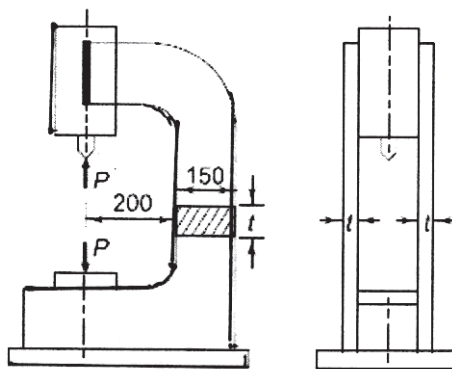


Figure 1.

OR

- Q2)** The layout of a transmission shaft carrying two pulleys A and B is shown in Fig. 2. Power is supplied to the shaft by means of a vertical belt on the pulley A, which is then transmitted to the pulley B carrying a horizontal belt. The ratio of belt tension on tight and slack side is 3:1. The limiting value of tension in the belts is 2.7 kN. The shaft is made of plain carbon steel 40C8 ($S_{ut} = 650 \text{ N/mm}^2$ and $S_{yt} = 380 \text{ N/mm}^2$) The pulleys are keyed to the shaft. Determine the shaft diameter according to the ASME code if $k_b = 1.5$ and $k_t = 1$. **[10]**

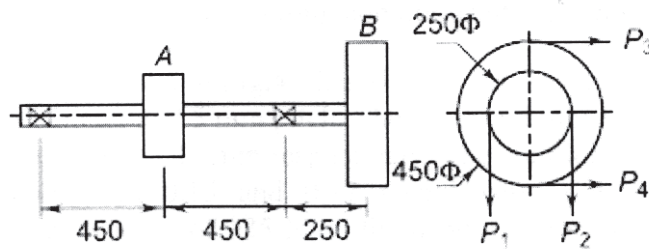


Figure 2

- Q3)** A cantilever beam made of cold drawn steel 40C8 ($S_{ut} = 600 \text{ N/mm}^2$ and $S_{yt} = 380 \text{ N/mm}^2$) is shown in Fig. 3. The force 'P' acting at the free end varies from -50 N to $+150 \text{ N}$. The expected reliability is 90% and the factor of safety is 2. The notch sensitivity factor at the fillet is 0.9. Determine the diameter 'd' of the beam at the fillet cross-section. Assume surface finish, size, reliability and theoretical stress concentration factors are 0.77, 0.85, 0.897 and 1.44 respectively. **[10]**

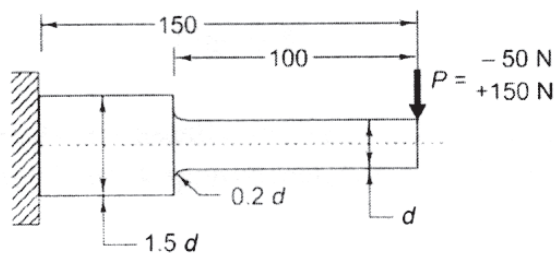


Figure 3

OR

- Q4) a)** Derive stress equations for flat key with suitable figure. [4]
- b)** The work cycle of a mechanical component subjected to completely reversed bending stresses consists of the following three elements: [6]
- $\pm 350 \text{ N/mm}^2$ for 85% of time
 - $\pm 400 \text{ N/mm}^2$ for 12% of time
 - $\pm 500 \text{ N/mm}^2$ for 3% of time
- The material for the component is 50C4 ($S_{ut}=660 \text{ N/mm}^2$) and the corrected endurance limit of the component is 280 N/mm^2 . Determine the life of the component.

- Q5) a)** Derive an expression for maximum efficiency of the square threaded screw. [6]
- b)** A double-threaded power screw, with ISO metric trapezoidal threads is used to raise a load of 300 kN. The nominal diameter is 100 mm and the pitch is 12mm. The coefficient of friction at the screw threads is 0.15. Neglecting collar friction, calculate; [12]
- torque required to raise the load;
 - torque required to lower the load; and
 - efficiency of the screw.

OR

- Q6) a)** Explain self-locking and overhauling conditions of the screw. [6]
- b)** A machine vice as shown in Fig.4 has single start, square threads with 22 mm nominal diameter and 5 mm pitch. The outer and inner diameters of the friction collar are 55 and 45 mm respectively. The coefficient of friction for thread and collar are 0.15 and 0.17 respectively. The machinist can comfortably exert a force of 125 N on the handle at a mean radius of 150 mm. Assuming uniform wear for the collar, calculate: [12]
- the clamping force developed between the jaws; and
 - the overall efficiency of the clamp

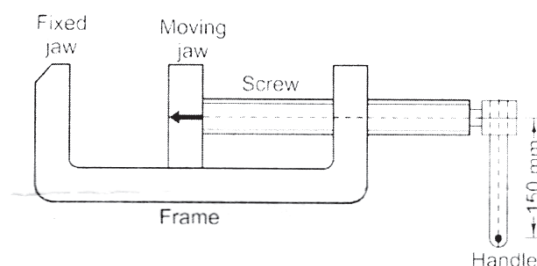


Figure 4

- Q7) a)** Prove that, the threads are equally strong in failure by shear and failure by tension if the height of nut is approximately 0.4 times the nominal diameter of bolt [4]
- b)** The structural connection shown in Fig. 5 is subjected to an eccentric force P of 10 kN with an eccentricity of 500 mm from the CG of the bolts. The centre distance between bolts I and 2 is 200 mm, and the centre distance between bolts I and 3 is 150 mm. All the bolts are identical. The bolts are made from plain carbon steel 30C8 ($S_{yt} = 400 \text{ N/mm}^2$) and the factor of safety is 2.5. Determine the size of the bolts. [12]

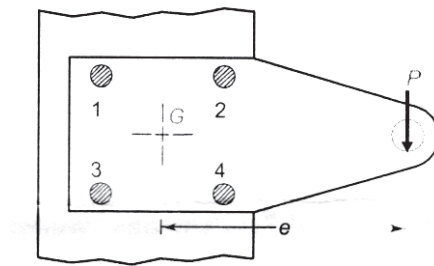


Figure 5

OR

- Q8) a)** Explain any two types of fillet joints with suitable sketch [4]
- b)** A steel plate, 100 mm wide and 10 mm thick, is joined with another steel plate by means of single transverse and double parallel fillet welds, as shown in Fig. 6. The strength of the welded joint should be equal to the strength of the plates to be joined. The permissible tensile and shear stresses for the weld material and the plates are 70 and 50 N/mm^2 respectively. Find the length of each parallel fillet weld. Assume the tensile force acting on the plates as static. [12]

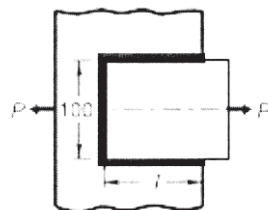


Figure 6

- Q9) a)** Explain various types of end styles for helical compression spring. **[4]**
- b)** A helical compression spring, made of circular wire, is subjected to an axial force, which varies from 2.5 kN to 3.5 kN. Over this range of force, the deflection of the spring should be approximately 5 mm. The spring index can be taken as 5. The spring has square and ground ends. The spring is made of patented and cold-drawn steel wire with ultimate tensile strength of 1050 N/mm² and modulus of rigidity of 81370 N/mm². The permissible shear stress for the spring wire should be taken as 50% of the ultimate tensile strength. Design the spring and calculate: **[12]**
- i) Wire diameter;
 - ii) Mean coil diameter;
 - iii) Number of active coils;
 - iv) Total number of coils;
 - v) Solid length of the spring;
 - vi) Free length of the spring;
 - vii) Required spring rate; and
 - viii) Actual spring rate

OR

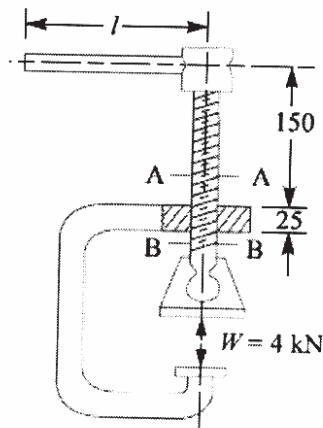
- Q10)a)** Explain with neat sketch nipping of leaf spring **[4]**
- b)** A concentric spring is used as a valve spring in a heavy duty diesel engine. It consists of two helical compression springs having the same free length and same solid length. The composite spring is subjected to a maximum force of 6000 N and the corresponding deflection is 50 mm. The maximum torsional shear stress induced in each spring is 800 N/mm². The spring index of each spring is 6. Assume same material for two springs and the modulus of rigidity of spring material is 81370 N/mm². The diametral clearance between the coils is equal to the difference between their wire diameters. Calculate: **[12]**
- i) The axial force transmitted by each spring;
 - ii) Wire and mean coil diameters of each spring;
 - iii) Number of active coils in each spring.



[6003]-422**T.E. (Mechanical /Mechanical Sandwich)****DESIGN OF MACHINE ELEMENTS****(2019 Pattern) (Semester-I) (302043)****Time : 2½ Hours]****[Max. Marks : 70****Instructions to the candidates:**

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1) a)** A C-clamp, as shown in Figure, has trapezoidal threads of 12 mm outside diameter and 2 mm pitch. The coefficient of friction for screw threads is 0.12 and for the collar is 0.25. The mean radius of the collar is 6 mm. If the force exerted by the operator at the end of the handle is 80 N, find:
- i) The length of handle;
 - ii) The maximum shear stress in the body of the screw and where does this exist; and
 - iii) The bearing pressure on the threads.
- [8]**



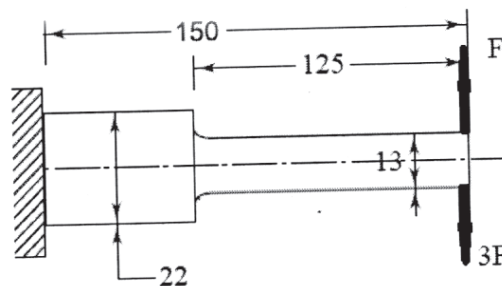
- b) Derive expression for torque required to lower the loads in case of square threads. **[5]**
- c) What are the different types of screw threads used for power screw? **[4]**
What are the advantages and limitations of power screw?

OR**P.T.O.**

- Q2) a)** A vertical two start square threaded screw of 100 mm mean diameter and 20 mm pitch supports a vertical load of 18 kN. The nut of the screw is fitted in the hub of a gear wheel having 80 teeth which meshes with a pinion of 20 teeth. The mechanical efficiency of the pinion and gear wheel drive is 90 percent. The axial thrust on the screw is taken by a collar bearing 250 mm outside diameter and 100 mm inside diameter. Assuming uniform pressure conditions, find, minimum diameter of pinion shaft and height of nut, when coefficient of friction for the vertical screw and nut is 0.15 and that for the collar bearing is 0.20. The permissible shear stress in the shaft material is 56 MPa and allowable bearing pressure is 1.4 N/mm². **[8]**
- b)** Explain self-locking and over-hauling property of screw. Prove the condition for screw to be self-locking. **[5]**
- c)** Prove that efficiency of self-locking square threads is less than 50%. **[4]**
- Q3) a)** A forged steel bar, 50 mm in diameter, is subjected to a reversed bending stress of 250 N/mm². The bar is made of steel 40C8 ($S_{ut}=600$ N/mm²). Calculate the life of the bar for a reliability of 90%. Take surface finish factor 0.44, Size factor 0.85, Reliability factor 0.897. **[8]**
- b)** What is modifying factor to account for stress concentration? Explain Endurance strength modifying factors? **[5]**
- c)** Explain with neat sketch the Gerber curve, Soderberg and Goodman lines? **[5]**

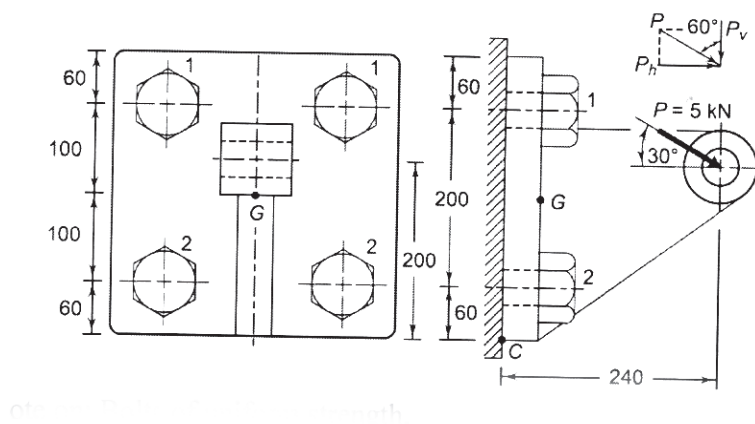
OR

- Q4) a)** A cantilever beam made of steel material with $S_{ut}=550$ N/mm², $S_{yt}=320$ N/mm² as shown in figure is subjected to load which varies from $-F$ to $3F$. Take surface finish factor 0.89, Size factor 0.85, Theoretical stress concentration factor 1.42, notch sensitivity 0.9, factor of safety 2. Determine maximum value of load F which the cantilever beam can withstand for infinite life. **[8]**



- b) Explain modified Goodman diagram. Draw neat labeled sketches of modified Goodman diagram for axial & Bending stresses. [5]
- c) Explain Fatigue Design under Combined Stresses. [5]

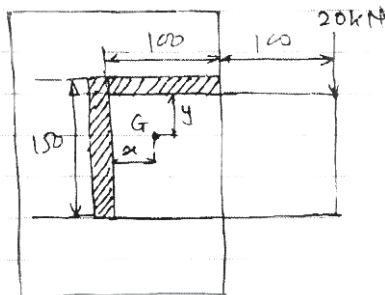
Q5) a) A bracket, subjected to a force of 5 kN inclined at an angle of 60° with the vertical, is shown in figure. The bracket is fastened by means of four identical bolts to the structure. The bolts are made of plain carbon steel 30C8 ($S_{yt} = 400 \text{ N/mm}^2$) and the factor of safety is 5 based on maximum shear stress. Assume maximum shear stress theory and determine the size of the bolts. [8]



- b) Write a note on: Bolts of uniform strength. [5]
- c) Explain the procedure for the design of bolts for eccentrically loaded bolted joints in shear. [5]

OR

Q6) a) Figure shows a welded joint subjected to a load of 20 kN. Find size of weld if permissible shear stress 80 MPa. [10]



Total No. of Questions : 8]

SEAT No. :

PA-1530

[Total No. of Pages : 4

[5926] - 150

T.E. (Mechanical Sandwich Engg.)

DESIGN OF MACHINE ELEMENTS

(2019 Pattern) (Semester - I) (302043)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) *Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Assume suitable data, if necessary.*

UNIT - I

- Q1) a)** The lead screw of a lathe has Acme threads of 50 mm outside diameter and 8 mm pitch. The screw must exert an axial pressure of 2500 N in order to drive the tool carriage. The thrust is carried on a collar 110 mm outside diameter and 55 mm inside diameter and the lead screw rotates at 30 r.p.m. Determine **[8]**
- i) the power required to drive the screw; and
 - ii) the efficiency of the lead screw. Assume a coefficient of friction of 0.15 for the screw and 0.12 for the collar. Also state the condition of screw.
- b) Derive expression for torque required to raise the loads in case of square threads. **[5]**
- c) Derive and expression for maximum efficiency of square threads. **[4]**

OR

- Q2) a)** A power screw having double start square threads of 25 mm nominal diameter and 5 mm pitch is acted upon by an axial load of 10 kN. The outer and inner diameters of screw collar are 50 mm and 20 mm respectively. The coefficient of thread friction and collar friction may be assumed as 0.2 and 0.15 respectively. The screw rotates at 12 r.p.m. Assuming uniform wear condition at the collar and allowable thread bearing pressure of 5.8 N/mm², find: 1. the torque required to rotate the screw; 2. the stress in the screw; and 3. the number of threads of nut in engagement with screw. Also state the condition of screw. **[8]**

P.T.O.

- b) What are the advantages of trapezoidal threads over square threads? State the meaning of each term involved in designation T r40 × 14(p7). [5]
- c) Explain self-locking and over-hauling property of screw. Prove the condition for screw to be self-locking. [4]

UNIT - II

- Q3)** a) A rotating bar made of steel 45C8 ($S_{ut} = 630$ MPa) is subjected to a completely reversed bending stress. The corrected endurance limit of the bar is 315 N/mm^2 . Calculate the fatigue strength of the bar for a life of 90,000 cycles. [8]
- b) What is stress concentration? What are the causes of stress concentration? [5]
- c) Explain with neat sketch the Gerber curve, Soderberg and Goodman lines? [5]

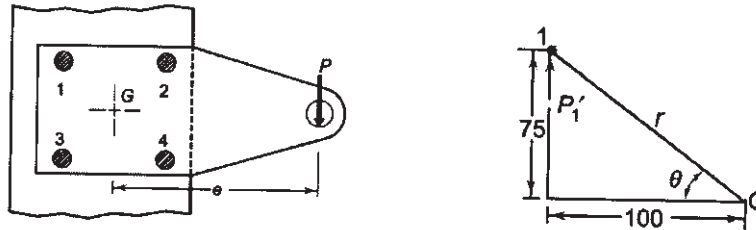
OR

- Q4)** a) A machine component is subjected to fluctuating stress that varies from 40 to 100 N/mm^2 . The corrected endurance limit stress for the machine component is 270 N/mm^2 . The ultimate tensile strength and yield strength of the material are 600 and 450 N/mm^2 respectively. Find the factor of safety using [8]
- i) Gerber theory
 - ii) Soderberg line
 - iii) Goodman line. Also, find the factor of safety against static failure.
- b) What is modifying factor to account for stress concentration? Explain Endurance strength Modifying factors? [5]
- c) Explain : [5]
- i) Notch Sensitivity
 - ii) Fatigue life
 - iii) Endurance limit.
 - iv) Modified Goodman diagram.

UNIT - III

- Q5) a)** The structural connection shown in Figure is subjected to an eccentric force P of 10 kN with an eccentricity of 500 mm from the CG of the bolts. The centre distance between bolts 1 and 2 is 200 mm, and the centre distance between bolts 1 and 3 is 150 mm. All the bolts are identical. The bolts are made from plain carbon steel 30C8 ($S_{yt} = 400 \text{ N/mm}^2$) and the factor of safety is 2.5. Determine nominal diameter of the bolts.

[8]

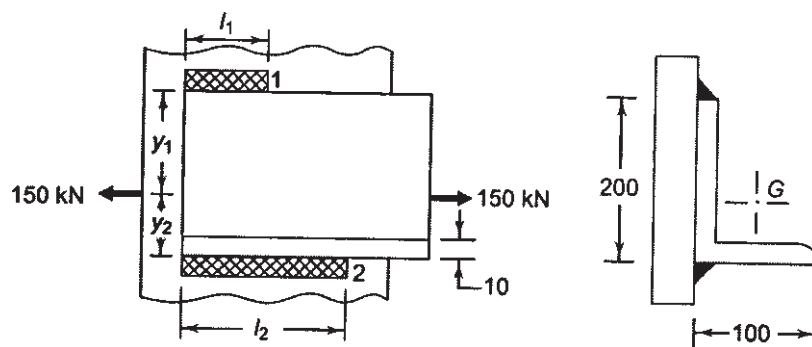


- b) Write a note on : Bolts of uniform strength. [5]
 c) Discuss the advantages and disadvantages of welded joints. [5]

OR

- Q6) a)** An ISA $200 \times 100 \times 10$ angle is welded to a steel plate by means of fillet welds as shown in Figure. The angle is subjected to a static force of 150 kN and the permissible shear stress for the weld is 70 N/mm^2 . Determine the lengths of weld at the top and bottom.

[8]



- b) Discuss in brief strength of parallel fillet welds. [5]
 c) Prove that stress on the throat is equal to the ratio of force on weld to $0.707 \times s \times 1$. [5]

UNIT - IV

- Q7) a)** Design a close coiled helical compression spring for a service load ranging from 2250 N to 2750 N. The axial deflection of the spring for the load range is 6 mm. Assume a spring index of 5. The permissible shear stress intensity is 420 MPa and modulus of rigidity, $G = 84 \text{ kN/mm}^2$. Neglect the effect of stress concentration. Draw a fully dimensioned sketch of the spring, showing details of the finish of the end coils. [7]
- b) Explain the following terms for helical spring [6]
- i) Active and inactive coils
 - ii) Spring index
 - iii) Spring rate
- c) Explain with the neat sketch, nipping of leaf spring. [4]

OR

- Q8) a)** Design a helical compression spring for a pressure relief valve using following data; Operating pressure = 14.5 Bar; Valve lift at 18% pressure rise = 7 mm ; Diameter of valve = 37 mm ; Limiting mean coil diameter = 36 mm ; Maximum shear stress = 465 MPa ; $G = 85 \text{ GPa}$, Clash clearance = 15 % of Maximum deflection of spring, End style of spring is square and ground std. Take spring wire diameter 6, 6.5, 7, 7.5, 8, 8.5 (mm). [7]
- b) What is mean by spring surge and what is its effect? [5]
- c) Explain shot peening. [5]



[5870]-1022

T.E. (Mechanical/Mechanical Sandwich)
DESIGN OF MACHINE ELEMENTS
(2019 Pattern) (Semester - I) (302043)

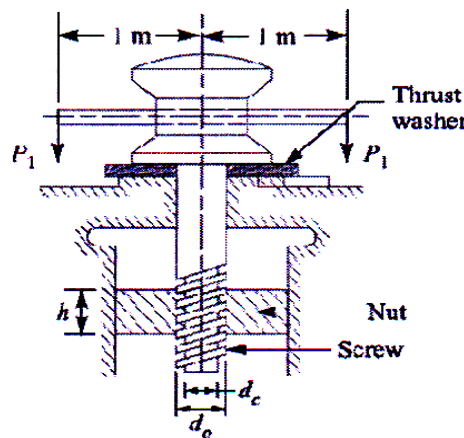
Time : 2½ Hours]

[Max. Marks : 70]

Instructions to the candidates:

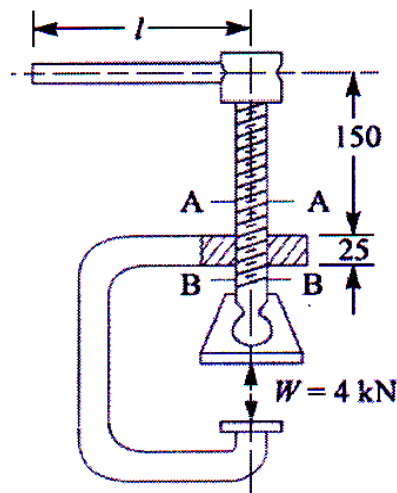
- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Draw neat labeled diagrams wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable/Standard data if necessary.
- 5) Use of non programmable electronic calculator is permitted.

- Q1)** a) What are the different types of screw threads used for power screw? [4]
 b) Derive the expression for the torque required to overcome collar friction. [5]
 c) A sluice gate weighing 18 kN is raised and lowered by means of square threaded screws, as shown in Fig. The frictional resistance induced by water pressure against the gate when it is in its lowest position is 4000 N. The outside diameter of the screw is 60 mm and pitch is 10 mm. The outside and inside diameter of washer is 150 mm and 50 mm respectively. The coefficient of friction between the screw and nut is 0.1 and for the washer and seat is 0.12. Find: i) The maximum force to be exerted at the ends of the lever raising and lowering the gate, ii) Efficiency of the arrangement, and iii) Number of threads and height of nut, for an allowable bearing pressure of 7 N/mm². [8]



OR

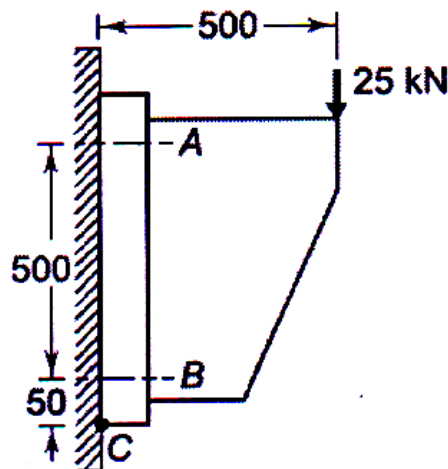
- Q2)** a) Describe the term: Virtual coefficients of friction. [4]
- b) Explain self-locking and over-hauling property of screw. Prove the condition for screw to be self-locking. [5]
- c) A C-clamp, as shown in Figure, has trapezoidal threads of 12 mm outside diameter and 2 mm pitch. The coefficient of friction for screw threads is 0.12 and for the collar is 0.25. The mean radius of the collar is 6 mm. If the force exerted by the operator at the end of the handle is 80 N, find: i) The length of handle; ii) The maximum shear stress in the body of the screw and where does this exist; and iii) The bearing pressure on the threads. [8]



- Q3)** a) Write a short note on; Modified Goodman diagram. [4]
- b) Write a short note on Fatigue Failure. [5]
- c) A spherical pressure vessel, with a 500 mm inner diameter, is welded from steel plates. The welded joints are sufficiently strong and do not weaken the vessel. The plates are made from cold drawn steel 20C8 ($S_{ut} = 440 \text{ N/mm}^2$ and $S_{yt} = 242 \text{ N/mm}^2$). The vessel is subjected to internal pressure, which varies from zero to 6 N/mm^2 . The expected reliability is 50% and the factor of safety is 3.5. The size factor is 0.85. The vessel is expected to withstand infinite number of stress cycles. Calculate the thickness of the plates. Take $K_a = 0.82$, $K_b = 0.85$, $K_c = 1$. [8]

OR

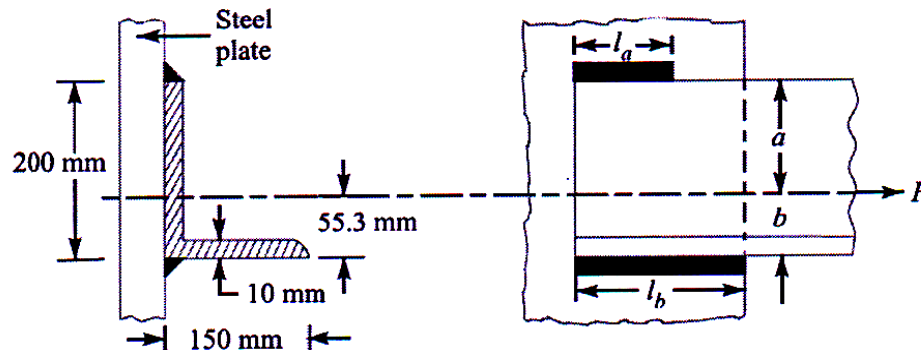
- Q4) a)** Define and explain Notch Sensitivity. [4]
- b) What is the difference between the Gerber curve, Soderberg and Goodman lines? [5]
- c) A machine component is subjected to fluctuating stress that varies from 40 to 100 N/mm². The corrected endurance limit stress for the machine component is 270 N/mm². The ultimate tensile strength and yield strength of the material are 600 and 450 N/mm² respectively. Find the factor of safety using (i) Gerber theory (ii) Soderberg line (iii) Goodman line Also, find the factor of safety against static failure. [8]
- Q5) a)** What are the assumptions made in the design of welded joint? [4]
- b) Discuss the procedure for designing an eccentric loaded welded joint. [6]
- c) A wall bracket is attached to the wall by means of four identical bolts, two at A and two at B, as shown in Figure. Assuming that the bracket is held against the wall and prevented from tipping about the point C by all four bolts and using an allowable tensile stress in the bolts as 35 N/mm², determine the size of the bolts on the basis of maximum principal stress theory. [8]



OR

- Q6) a)** State the advantages and disadvantages of welded joints. [4]
- b) Explain with neat sketch, following screw fasteners [6]
- i) Through Bolts ii) Tap Bolts iii) Cap Screws

- c) A $200 \times 150 \times 10$ mm angle is to be welded to a steel plate by fillet welds as shown in Figure. If the angle is subjected to a static load of 200 kN, find the length of weld at the top and bottom. The allowable shear stress for static loading may be taken as 75 MPa. [8]



- Q7) a) Define following terms : [4]
 i) Solid length
 ii) Spring index
 iii) Active and Inactive coils
 iv) Spring Rate
 b) Explain A.M. Wahl's factor and state its importance in the design of helical springs. [6]
 c) A helical spring is made from a wire of 6 mm diameter and has outside diameter of 75 mm. If the permissible shear stress is 350 MPa and modulus of rigidity 84 kN/mm^2 , find the axial load which the spring can carry and the deflection per active turn. [8]

OR

- Q8) a) How will you find whether the given helical spring is a compression spring or tension spring? [4]
 b) Explain with the neat sketch, nipping of leaf spring. [6]
 c) Design a valve spring of a petrol engine for the following operating conditions: Spring load when the valve is open = 400 N; Spring load when the valve is closed = 250 N; Maximum inside diameter of spring = 25 mm; Length of the spring when the valve is open = 40 mm; Length of the spring when the valve is closed = 50 mm; Maximum permissible shear stress 400 MPa. [8]

